K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

ONLINE ORDER PROCESSING AND TRACKING SYSTEM (OOP-TS

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CERTIFICATE

This is to certify that the project based laboratory report entitled "ONLINE ORDER PROCESSING AND TRACKING SYSTEM (OOP-TS)" submitted by **M.Hanumanth Reddy, Sanku. Sai , I.Nischal Sai, T.Shanker Sriteja** bearing Regd.No. 2500031811, 2500031871, 2500031935, 2500032062 to the **DEPARTMENT of BASIC ENGINEERING SICENCE-II, K L UNIVERSITY** in partial fulfillment of the requirements for the completion of a project based Laboratory in "Python Programming (25SC2007)" course in I B Tech II Semester, is a Bonafide record of the work carried out by them/him/her under my supervision during the academic year 2025 – 2026.

Dr. N SREERAM

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ABSTRACT

The Online Order Processing and Tracking System (OOP_TS) is designed to streamline and automate the process of placing, managing, and tracking customer orders in an efficient and user-friendly manner. This system leverages Object-Oriented Programming (OOP) principles such as encapsulation, inheritance, and polymorphism to ensure modularity, scalability, and maintainability.The application enables users to browse products, place orders, and track their order status in real-time. It also provides administrative functionalities such as managing inventory, updating order status, and handling customer details. The system ensures data accuracy, reduces manual effort, and enhances the overall user experience through automation.By implementing features like order confirmation, status updates (processing, shipped, delivered), and secure data handling, the system improves operational efficiency for businesses while offering transparency and convenience to customers. This project demonstrates how OOP concepts can be effectively applied to develop a robust and scalable real-world application.

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1.INTRODUCTION

In recent years, the growth of the internet and digital technologies has completely transformed the way businesses operate. Online shopping has become an essential part of everyday life, where customers can purchase products from anywhere at any time. Popular platforms like Amazon, Flipkart, and Meesho provide fast and efficient services by using advanced order processing and tracking systems.

An order processing system is the backbone of any e-commerce platform. It ensures that once a customer places an order, the system records the details, processes the request, updates inventory, and initiates delivery. Along with this, tracking systems allow customers and businesses to monitor the progress of orders from dispatch to final delivery. Without such systems, managing large numbers of orders would be extremely difficult and error-prone.

The **Online Order Processing and Tracking System (OOP-TS)** is developed as a simplified model of these real-world systems. It helps in understanding how orders are created, managed, and tracked using programming techniques. This project is implemented using Python and follows a structured approach using Object-Oriented Programming (OOP) concepts such as classes, objects, and methods. The system provides various functionalities including order placement, order status updates, searching orders, and tracking shipments. It also maintains records using file handling techniques, ensuring that data is stored and retrieved efficiently. In addition, the project introduces basic data analysis and visualization to study business performance.

This project is not only useful for learning programming concepts but also gives practical exposure to how real-time systems work in industries. It bridges the gap between theoretical knowledge and real-world applications, making it highly valuable for students and beginners

2. AIM

To design and develop a Python-based system that:

- Manages customer orders efficiently
- Tracks order delivery status
- Stores and retrieves order data
- Provides basic analytics and reports

2.1. ADVANTAGES

- Reduces manual work and errors
- Easy to use and understand
- Provides real-time order tracking
- Efficient data storage using files
- Helps in business analysis
- Demonstrates real-world application of Python

2.2. DISADVANTAGES

- Limited to small-scale usage
- No real-time online payment integration
- Basic user interface (console-based)
- Requires manual input
- Not suitable for large enterprise systems

2.3. FUTURE IMPLEMENTATION

- Add GUI using Tkinter or Web interface
- Integrate online payment systems (UPI, cards)
- Use database (MySQL) instead of JSON
- Add login authentication system

3. SYSTEM REQUIREMENTS

➢ SOFTWARE REQUIREMENTS:

The major software requirements of the project are as follows:
Language: Python.
Operating system**:** Windows 10 or more.
Technical requirements: Monitor, CPU, Keyboard, Mouse, etc...

➢ HARDWARE REQUIREMENTS:

• The hardware requirements that map towards the software are as follows:

RAM: 8 GB.

• Processor: i5/i7

4. ALGORITHM

Step-1: Start the program

Step-2: Display menu options

Step-3: Read user choice

Step-4: If choice = Place Order

Input customer details

Calculate total

Store order

Step-5 : If choice = View Orders

Display all orders

Step-6: If choice = Search Order

Input search value

Display matching results

Step-7: If choice = Update Status

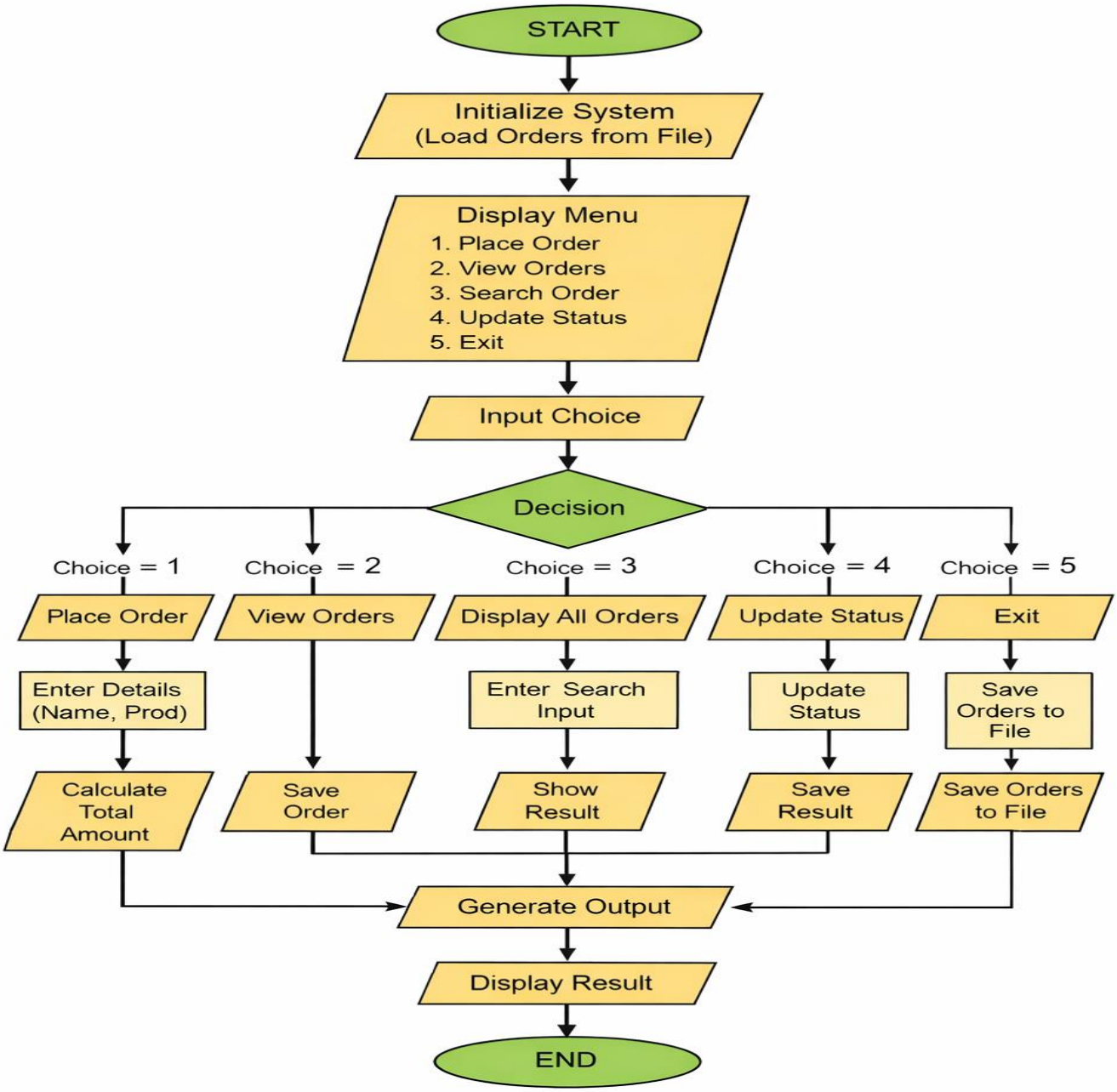
Input order ID

Update status

Step-8: Save data to file

Step-9: Exit program

5. FLOWCHART


START
Initialize System
(Load Orders from File)
Display Menu
1. Place Order
2. View Orders
3. Search Order
4. Update Status
5. Exit
Input Choice
Decision
Choice = 1
Choice = 2
Choice = 3
Choice = 4
Choice = 5
Place Order
View Orders
Display All Orders
Update Status
Exit
Enter Details
(Name, Prod)
Enter Search
Input
Update
Status
Save
Orders to
File
Calculate
Total
Amount
Save
Order
Show
Result
Save
Result
Save Orders
to File
Generate Output
Display Result
END

6. IMPLEMENTATION

```
import json


class Order:
    def __init__(self, order_id, customer, product, quantity, price):
        self.order_id = order_id
        self.customer = customer
        self.product = product
        self.quantity = quantity
        self.price = price
        self.status = "Placed"

    def calculate_total(self):
        return self.quantity * self.price

    def update_status(self, new_status):
        self.status = new_status

    def to_dict(self):
        return {
            "order_id": self.order_id,
            "customer": self.customer,
            "product": self.product,
            "quantity": self.quantity,
            "price": self.price,
            "status": self.status
        }

    def display(self):
        print(f"{self.order_id} | {self.customer} | {self.product} | "
              f"{self.quantity} | {self.price} | {self.status} | Total: {self.calculate_total()}")



def save_orders(orders):
    with open("orders.json", "w") as f:
        json.dump([o.to_dict() for o in orders], f, indent=4)


def load_orders():
    try:
        with open("orders.json", "r") as f:
            data = json.load(f)
```

```
        return [Order(**d) for d in data]
    except:
        return []


def add_order(orders):
    try:
        order_id = len(orders) + 1
        customer = input("Enter customer name: ")
        product = input("Enter product name: ")
        quantity = int(input("Enter quantity: "))
        price = float(input("Enter price: "))

        if quantity <= 0 or price <= 0:
            print("Invalid input!")
            return

        order = Order(order_id, customer, product, quantity, price)
        orders.append(order)

        print("✅ Order placed successfully!")

    except:
        print("❌ Error in input!")


def view_orders(orders):
    if not orders:
        print("No orders found!")
        return

    print("\nID | Customer | Product | Qty | Price | Status | Total")
    print("-" * 60)
    for o in orders:
        o.display()


def search_order(orders):
    name = input("Enter customer name to search:  ").lower()
    found = False

    for o in orders:
        if name in o.customer.lower():
            o.display()
            found = True
```

```
    if not found:
        print("No matching orders found!")


def update_order_status(orders):
    try:
        oid = int(input("Enter Order ID: "))
        for o in orders:
            if o.order_id == oid:
                print("1. Packed  2. Shipped  3. Delivered  4. Cancelled")
                choice = input("Select status: ")

                status_dict = {
                    "1": "Packed",
                    "2": "Shipped",
                    "3": "Delivered",
                    "4": "Cancelled"
                }

                if choice in status_dict:
                    o.update_status(status_dict[choice])
                    print("✅ Status updated!")
                else:
                    print("Invalid choice!")
                return

        print("Order not found!")

    except:
        print("Invalid input!")


def main():
    orders = load_orders()

    while True:
        print("\n===== ONLINE ORDER SYSTEM =====")
        print("1. Place Order")
        print("2. View Orders")
        print("3. Search Order")
        print("4. Update Status")
        print("5. Save & Exit")

        choice = input("Enter choice: ")
```

```
        if choice == "1":
            add_order(orders)

        elif choice == "2":
            view_orders(orders)

        elif choice == "3":
            search_order(orders)

        elif choice == "4":
            update_order_status(orders)

        elif choice == "5":
            save_orders(orders)
            print("Data saved. Exiting...")
            break

        else:
            print("Invalid choice!")


if __name__ == "__main__":
    main()
```

7. INPUT & OUTPUT SCREENSHOTS

Screen Shots:

```
=======  ONLINE ORDER SYSTEM  ======

1. Place Order
2. View Orders
3. Search Order
4. Update Status
5. Save & Exit

Enter choice: 1

Enter customer name: Rahul
Enter product name: Shoes
Enter quantity: 2
Enter price: 5000

✅ Order placed successfully!

Enter choice: 2

ID  | Customer | Product  | Qty   | Price | Status | Total
-------------------------------------------------------
1   | Rahul    | Shoes    | 2     | 500.0 | Placed |1000.0
-------------------------------------------------------

Enter choice: 2

ID | Customer | Product |
3                    Qty| Price |Status  | Total
-------------------------------------------
 1  | Rahul      | Shoes    | 2500.0| Placed  | 1000.0
```

8.CONCLUSION

The **Online Order Processing and Tracking System** successfully demonstrates how real-world e-commerce systems work. It provides a clear understanding of order management, tracking, and data handling using Python. The project enhances programming skills and helps in applying theoretical concepts to practical applications. It can be further improved and expanded into a full-scale application in the future.